PYTHON PROGRAMMING | EXERCISE 03

Morphological Analysis with NLTK

Lemmatizing inflected English words

This example uses WordNetLemmatizer to map inflected word forms to lemmas. The part of speech matters: the same surface form can be analyzed differently as a noun, verb, or adjective.

Question

Write a Python program that demonstrates morphological analysis using the NLTK library.

Code

```
import nltk
from nltk.stem import WordNetLemmatizer

# Install NLTK once with: python -m pip install nltk
# Download the small WordNet data set once if it is not already available.
try:
    nltk.data.find("corpora/wordnet")
except LookupError:
    if not nltk.download("wordnet", quiet=True):
        raise RuntimeError("Could not download the NLTK WordNet corpus.")

lemmatizer = WordNetLemmatizer()
items = [("cars", "n"), ("running", "v"), ("studies", "v"), ("better", "a")]

print("Word form | POS | lemma")
for word, pos in items:
    print(f"{word:9} | {pos}   | {lemmatizer.lemmatize(word, pos)}")
```

Output

```
Word form | POS | lemma
cars      | n   | car
running   | v   | run
studies   | v   | study
better    | a   | good
```

Output screenshot

